

STAT248 Lecture 2 Characteristics of Time Series

1 Characteristics of Time Series

1.1 Stochastic Process

- Stochastic process 为随着时间变化的 systems 或 phenomena
- Time series 可以被建模为 realizations of stochastic processes, 即便它们包括 deterministic / predictable 的 components

Logic ▾

大多数现实中的 time series 可以被视作由两部分组成:

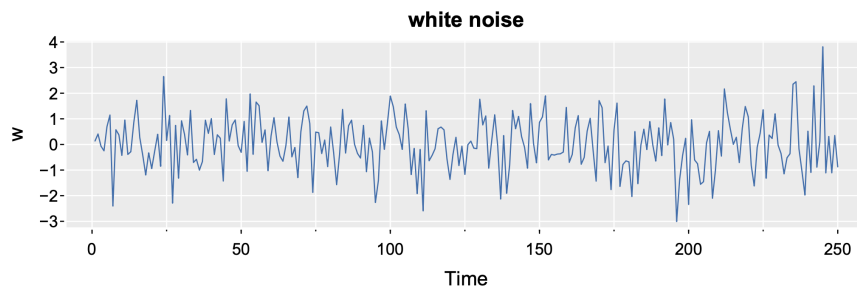
- underlying signal s_t
- noise w_t

在某些 (好的) 特殊情况下, w_t 为 white noise

1.2 Definition: White Noise

White noise 是一个 uncorrelated variables in sequence $\{x_t\}_{t=1,2,\dots}$ 的特殊情况, 对任意 $t \in \{1, 2, \dots\}$, 其满足:

- $x_t \stackrel{i.i.d.}{\sim}$ some distribution f
- $\mathbb{E}[x_i] = 0$
- $\text{Var}[x_i] =: \sigma_w^2$



Example: Gaussian White Noise ▾

一种常见且有用的 white noise 取自 Gaussian distribution, 即

$$w_t \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_w^2), \quad t = 1, 2, \dots$$



Python

```
1 def white_noise(nt, nobs=1, var=1):
2     '''
3     Generate a time series with uncorrelated random variables, 'w_t',
4     with mean 0 and finite variance 'var'.
5     If var=1 this is the standard normal distribution.
6     Inputs:
7         nt [int] : number of time points
8         nobs [int] : number of white noise time series to generate
9         var [float] : variance
10    Output:
```

```

11     w [np.array] = array of length nt
12     ...
13     w = np.sqrt(var) * np.random.randn(nt,nobs)
14     return w

```

Logic ▾

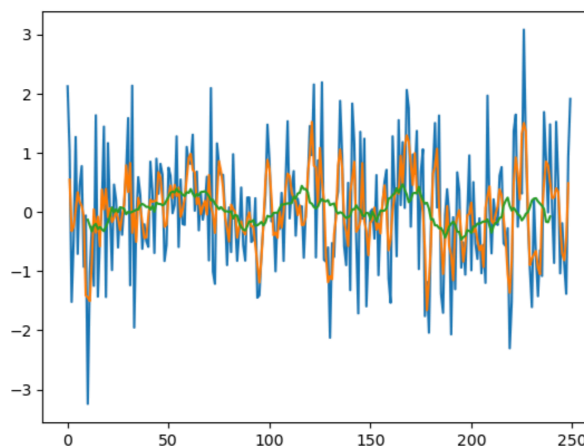
一种 smoothing a time series (包括 white noise) 的方式是使用 moving average

1.3 Definition: Moving Average

Moving average (MA) 会将时间点 t 与其邻近的 window 中的值进行平均

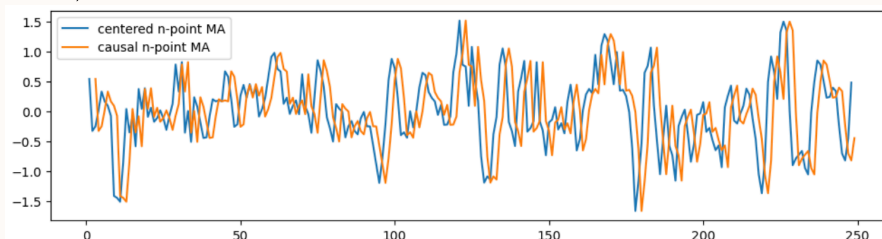
令 window n 为 odd, 则 moving average 被定义为:

$$v_t = \frac{1}{n} \sum_{k=-\frac{n-1}{2}}^{\frac{n-1}{2}} w_{t+k}$$



Remark ▾

- Moving average 可以视作一个 **Low pass filter** (保留低频信号, 抑制高频信号), 其可以:
 - 保留周期大于 n samples 的慢速趋势
 - 抑制周期小于约 n samples 的快速振荡
- 注意到 moving average 会涉及 "未来" 的数据, 若仅涉及 "过去" 的数据, 则被称为 **causal MA**, 例如 $v_t = \frac{1}{3}(x_{t-2} + x_{t-1} + x_t)$



- 我们通常使用 moving average 来进行简单的在线平滑, 以揭示 noisy time series data 的趋势, 并消除 noise 带来的波动
- Moving average 是最基本的平滑方式, 我们通常采用更复杂的方式, 如 exponential moving averages, Kalman filters, median filter

Example: 3-point MA ▾

若 window 为 3, 则

$$v_t = \frac{1}{3}(x_{t-1} + x_t + x_{t+1})$$



Python

```
1 def moving_average(w, n=3):
2     '''
3     Create an acausal moving average of the time series `w`
4     using an `n`-point moving average (default 3).
5     '''
6     w = np.asarray(w)
7     if w.ndim != 2:
8         w = np.atleast_2d(w).T
9     nt, nob = w.shape
10    v = np.zeros((nt, nob)) * np.nan
11    half_win = n // 2
12    for t in np.arange(half_win, nt - half_win):
13        v[t,:] = np.mean(w[t-half_win : t+half_win+1,:], axis=0)
14    return v
```

Logic ▾

一种常见的包含 white noise 的数据集来自 autoregressive process

1.4 Definition: Autoregression

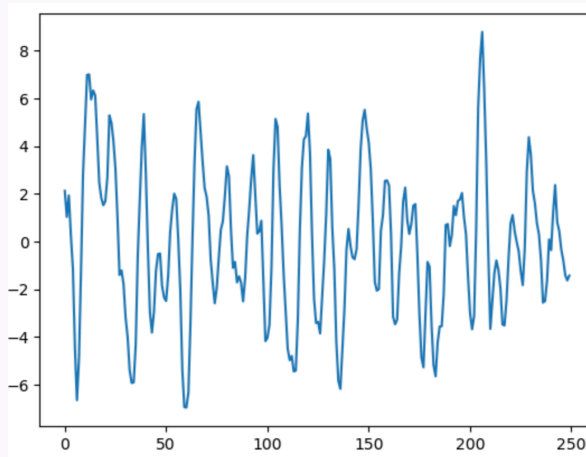
Autoregression 表示基于同一个 time series 的 past values 所进行的 regression / prediction

Example: AR(2) Process ▾

当前时刻的 data 取决于前 2 个时刻的 data

$$x_t = 1.5x_{t-1} - 0.75x_{t-2} + w_t$$

注意到 t 处的数据依赖于 $t-1$ 和 $t-2$, 因此该 process 为一个 AR(2) process



Logic ▾

另一个简单但重要的 extension to the idea of white noise 是 random walk

1.5 Definition: random walk

Random walk 被定义为:

- $x_0 = 0$
- $x_t = x_{t-1} + w_t$

另一种等价写法是:

$$x_t = \sum_{j=1}^t w_j$$

Random walk 在时刻 t 的 mean:

$$\mathbb{E}[x_t] = \sum_{j=1}^t \mathbb{E}[w_j] = 0$$

Random walk 在时刻 t 的 variance:

$$\text{Var}[x_t] = \text{Var}\left[\sum_{j=1}^t w_j\right] = t\sigma^2$$

因此方差会随着时间线性增长

⚠ Remark ▾

- Random walk 的数据之间不独立
- Random walk 的数据之间不同分布, 因为方差会随时间变化

☰ Example ▾

- Stock prices (short time scales)
- Brownian motion (continuous time)
- Null models for decision making (no evidence accumulation)



Python

```
1 def random_walk(nt, x0=0):
2     '''
3     Create a random walk for `nt` time points and with
4     initial value `x0`.
5     Inputs:
6         nt (int) : number of time points
7         x0 (float) : initial value
8     '''
9     x = np.zeros((nt,))
10    x[0] = x0
11    for n in np.arange(1,nt):
12        x[n] = x[n-1] + white_noise(1).item()
13    return x
```

🔄 Logic ▾

相较于 random walk, 更常见的情况是 random walk with drift

1.6 Definition: random walk with drift

Random walk with drift δ 被定义为:

- $x_0 = 0$
- $x_t = \delta + x_{t-1} + w_t$

另一种等价写法是:

$$x_t = \delta t + \sum_{j=1}^t w_j$$

Random walk with drift 在时刻 t 的 mean:

$$\mathbb{E}[x_t] = \delta t + \sum_{j=1}^t \mathbb{E}[w_j] = \delta t$$

因此均值会随着时间线性增长

⚠ Remark ▾

注意到此处 expected value 与 drift term 直接相关, 因此若我们知道 drift value, 则可以得到条件期望:

$$\mathbb{E}[x_t | x_s] = x_s + \delta(t - s)$$

Random walk with drift 在时刻 t 的 variance:

$$\text{Var}[X_t] = \text{Var}\left[\sum_{j=1}^t w_j\right] = t\sigma^2$$

因此方差会随着时间线性增长

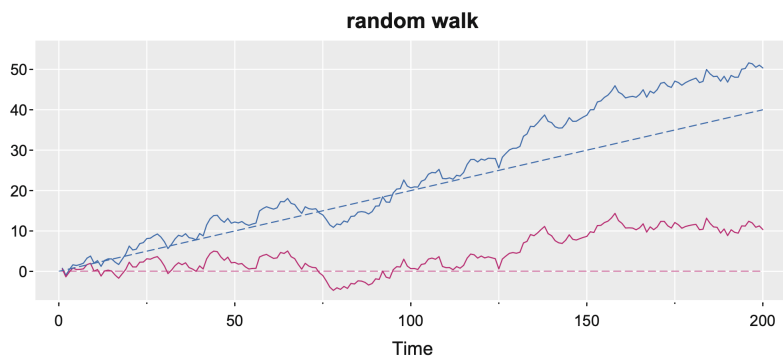


Fig. 1.11. Random walk, $\sigma_w = 1$, with drift $\delta = .2$ (upper jagged line), without drift, $\delta = 0$ (lower jagged line), and straight (dashed) lines with slope δ

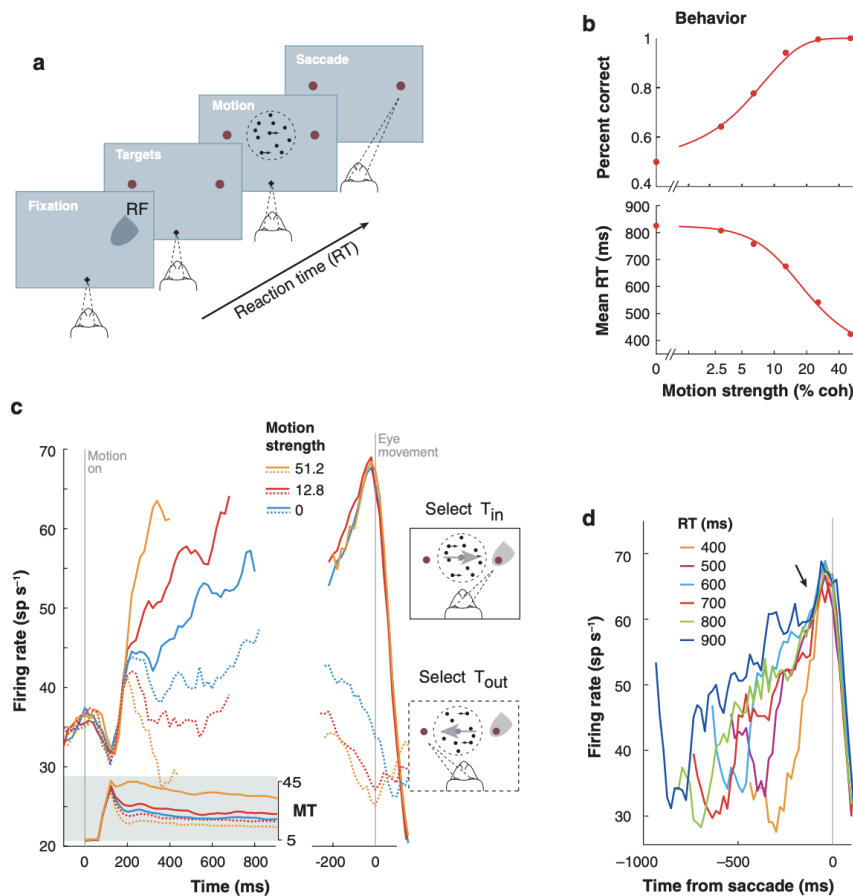
☰ Example ▾

- Stock prices (large time scales)
- Genetic drift
- Atmosphere CO_2 concentration

☰ Example: Drift Diffusion Models (漂移扩散模型) ▾

Drift diffusion models 是一个 cognitive model, 用于解释 evidence accumulation in decision making

- x_t 表示 decision variable
- δ 表示 mean evidence accumulation at each time step
- A 表示 boundary condition
- RT 表示 reaction time (x_t 第一次越过 A 或 $-A$ 的时刻)



Python

```
1 def random_walk_drift(nt, drift, x0=0):
2     ...
3     Create a random walk with drift for `nt` time points, `drift` drift, and
4     with initial value `x0`.
5     Inputs:
6         nt (int) : number of time points
7         drift (float) : constant drift
8         x0 (float) : initial value
9     ...
10    x = np.zeros((nt,))
11    x[0] = x0
12    for n in np.arange(1,nt):
13        x[n] = x[n-1] + white_noise(1).item() + drift
14    return x
```

Logic

更一般地, 数据常来自于被 white noise 污染的 periodic signals

1.7 Definition: Signal Plus Noise

一个常见的例子是 sinusoid:

$$x_t = A \cos(2\pi\omega t + \phi) + w_t$$

其中,

- A 表示 amplitude of the signal
- ω 表示 frequency of the oscillation
- ϕ 表示 phase shift

其中 signal amplitude 与 noise standard deviation 的比值 (此处为 $\frac{A}{\sigma_w}$) 被定义为 **signal to noise ratio (SNR)**, 信噪比越大, 恢复信号就越容易